Sources:

Vitamin B_{12} is found in animal but not in vegetable foods and is unique in that it is the only known vitamin that holds an ion of metal (cobalt) in its molecule. Eggs and meat supply ample amounts of this vitamin.

Functions:

- Many microorganisms require vitamin B_{12} for growth.
- Vitamin B_{12} is required as coenzyme for the conversion of L-methylmalonyl CoA to Succinyl CoA by the enzyme methylmelonyl CoA mutase. This reaction is essential for in the metabolisom of some branched-chain amino acids and odd-chain fatty acids
- Vitamin B_{12} is required for the maturation of red blood cells in the bone marrow and for the synthesis of proteins.

Absorption and Storage

Vitamin B_{12} is absorbed from ileum. For the absorption of vitamin B_{12} from the intestines, a factor called "Intrinsic Factor" (IF) secreted by the stomach is essential. Vitamin B_{12} is stored in fair amounts in the liver.

Deficiency:

When absorption is prevented by lack of intrinsic factor, it leads to the condition called pernicious anemia. It is characterized by a drastic decrease in red blood cell count and leads to formation of macrocytic red blood cells.

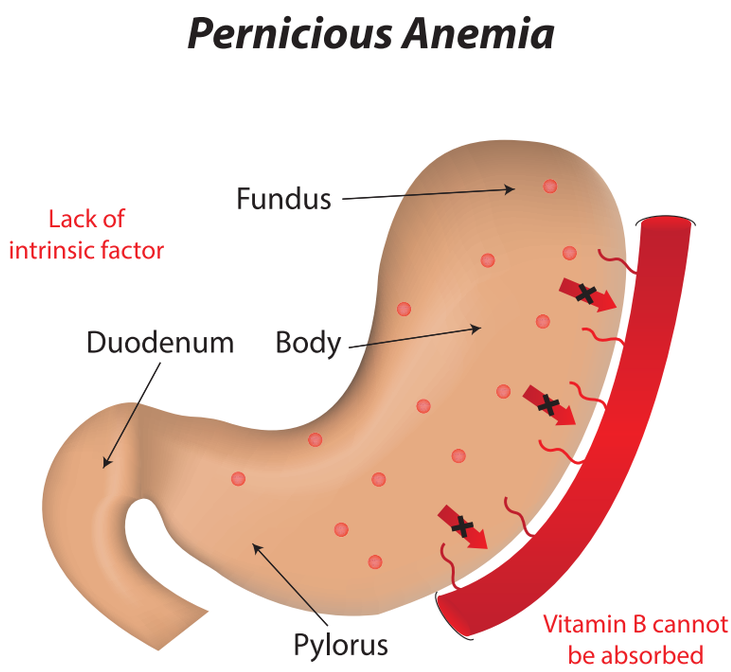


Figure 8.24 Pernicious Anemia

Vitamin B_{12} deficiency causes an increased concentration of methyl malonic acid, which competes with malonyl CoA and impairs fatty acid synthesis.

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